

Complementary choices of variables, representation and coupling; not mutually exclusive model classes.

E Cell–environment coupling

Cells + chemical field



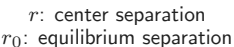
Cell senses local gradient

Arrow: gradient direction
Contours: constant c

Cells + fiber network

Possible coupling
Dashed link: dependency, not force

Dots: cells; $\int K_h \, d^2x = 1$



Coarse-graining

Which degrees of freedom are averaged out?

Illustrative geometry and prescribed analytic fields; not to scale, not simulation results. No panel ordering by resolution.